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The Mandatory Edge: A Technical Blueprint for Integrating Gaze and Pupil Response in a Fail-Closed AI Interview Proctoring System

Architectural Foundation for Mandatory Compliance

The implementation of a mandatory dual-modal biometric system hinges on an architectural design that makes proctoring an inseparable and non-disruptable component of the interview experience itself. This is not achieved through simple software flags but through a deeply integrated, fail-closed framework where the proctoring logic runs on the client-side, directly within the browser environment [499]. The core of this enforcement strategy lies in leveraging modern web technologies, specifically WebAssembly (Wasm) and WebGPU, to execute complex computer vision (CV) models locally [59] [104]. This approach ensures that any attempt to circumvent the system—by closing the browser tab, switching applications, or disabling the camera—results in an immediate and definitive halt to the interview session. The system's mandatoriness is thus enforced by design, making the proctoring engine a fundamental part of the application's runtime state. The initial Product Requirements Document (PRD) correctly identifies the need for such a system, outlining functional requirements like continuous passive eye tracking and active liveness challenges, while the Technical Requirements Document (TRD) specifies the underlying technology stack necessary to achieve them .

The primary technical enabler for this mandatory architecture is the use of WebAssembly and WebGPU for client-side inference [61] [411]. These technologies allow developers to compile high-performance machine learning models, originally written in languages like C++ or Python, into a format that can run at near-native speed inside a web browser [230] [411]. The TRD mandates that real-time CV inference must operate with less than 200ms latency on the client side, a performance target that was once unattainable in the browser but is now feasible due to the computational power offered by WebGPU . Benchmarks show that WebGPU can accelerate ML workloads by over 30 times compared to CPU-based Wasm execution, enabling the processing of high-resolution video streams from a standard webcam at 30 frames per second (FPS) or more [102] [103] [410]. This local processing

capability is the cornerstone of the system's integrity. The entire proctoring engine, including models for face landmark detection, gaze estimation, and pupil diameter measurement, is bundled with the web application package and executed within the client's sandboxed environment [86] [91]. Because this engine is integral to the application's function, its failure directly impacts the user experience. If the engine crashes, becomes unresponsive, or is terminated, the application's main thread can detect this state and trigger a predefined "fail-closed" action, such as pausing the interview, displaying a critical error message, and logging the event [499]. This creates a powerful deterrent; a candidate attempting to disable the proctoring would find the interview session itself frozen, making circumvention futile. This principle is further reinforced by a fallback mechanism described in the TRD, which alerts the user to close background applications if FPS drops below a threshold of 15 FPS, demonstrating a multi-layered approach to maintaining system health and operational continuity .

The system's compliance is further hardened by a "fail-closed" design philosophy, which dictates that in any state of uncertainty or detected anomaly, the system defaults to a secure, locked-down state rather than a permissive one [499]. This extends beyond just the health of the CV engine. The architecture must account for various points of potential failure or attack. For instance, if the client's webcam feed is lost, the network connection degrades below a minimum upload speed of 5 Mbps, or the jitter exceeds 100ms, the system must halt the session to prevent a loss of monitoring integrity . Similarly, the system must actively monitor for candidate actions that could compromise the test environment. The PRD explicitly calls for the detection of secondary monitors, mobile phones, and unauthorized browser tabs/extensions . When such an entity is detected via object detection models running on the client side (e.g., YOLOv8), the system can follow a graduated response protocol: first issuing a gentle UI warning, then temporarily locking the screen, and finally terminating the session if the behavior persists. This automated, rule-based enforcement ensures that the proctoring system is not merely observing passively but is actively participating in maintaining the controlled environment. The entire frontend application, from the initial hardware check and environmental scan to the continuous monitoring during the interview, is designed as a closed-loop system where every component feeds back into the central proctoring logic. The UI itself acts as an enforcement tool, guiding the candidate through mandatory steps like the initial calibration and providing clear instructions when deviations occur, thereby reducing ambiguity and reinforcing the system's non-negotiable nature .

The following table outlines the key components of the mandatory architectural foundation, detailing their functions and the technologies that enable them.

Component	Function	Enabling Technology / Protocol
Client-Side Inference Engine	Executes all CV models (face detection, gaze estimation, pupil tracking) locally on the candidate's device.	WebAssembly (Wasm), TensorFlow.js, ONNX Runtime Web [61] [91] [105]
WebGPU Acceleration	Provides GPU-level parallel processing for ML models, achieving low-latency (<200ms) inference on consumer hardware.	WebGPU API, WGSL (Shading Language) [64] [92] [320]
Fail-Closed Enforcement	Halts the interview session and logs an event upon detection of system failures (e.g., engine crash, low FPS) or candidate circumvention attempts.	Application-level state management, WebSocket error handling [499]
Hardware & Environment Check	Verifies webcam functionality, lighting conditions, and scans the room for unauthorized devices before the session starts.	WebRTC, MediaPipe Pose, YOLOv8 Object Detection
Real-Time Telemetry Streaming	Continuously sends processed biometric data (gaze, pupil size, head pose) from the client to the cloud backend.	WebSockets (`/ws/v1/telemetry` endpoint) with compressed payloads (MessagePack/ Protobuf) [252]

This architectural blueprint transforms the concept of "mandatory" from a policy statement into a technical reality. By embedding the proctoring logic deep within the client-side application and leveraging the performance of WebGPU, the system creates a resilient and difficult-to-bypass monitoring environment. Every element, from the choice of WebAssembly for portability and security to the design of a fail-closed state machine, is oriented towards ensuring that the dual-modal gaze and pupil monitoring is not just a feature, but a foundational, non-negotiable aspect of the AI Interview Room platform. The implementation plan detailed in the provided documents reflects this by allocating significant resources to Computer Vision engineers focused on edge inference, underscoring its strategic importance .

Implementation of Real-Time Gaze Tracking

The implementation of real-time gaze tracking is a multi-stage process that transforms raw video data from a standard webcam into a continuous stream of precise, actionable data about a candidate's point of regard [155]. This modality serves as the behavioral layer of the dual-system, capturing voluntary eye movements indicative of attention, distraction, or deception. The technical workflow, as synthesized from extensive research, involves a sequence of well-defined computer vision and machine learning tasks, each building upon the last to produce a robust gaze estimation model [4] [509]. The ultimate goal is to meet the stringent accuracy and latency requirements outlined in the TRD, ensuring the data is both reliable and usable in real-time without disrupting the interview flow . The system must be capable of operating effectively under typical home office

conditions, which presents challenges related to ambient lighting, head movement, and varying webcam quality [428].

The foundational step in any gaze tracking pipeline is the accurate detection and modeling of the human face and eyes [4]. This is typically achieved using pre-trained deep learning models optimized for speed and accuracy on consumer-grade hardware. The MediaPipe Face Mesh solution is a prime example, capable of detecting 468 landmarks on the face in real-time using a standard RGB webcam [4] [91]. These landmarks provide a stable geometric reference frame, precisely locating the boundaries of the eyes, nose, and mouth. From this, the system can isolate the region of interest—the eyes—which is crucial for the subsequent stages of pupil detection and gaze calculation. This initial detection phase is computationally intensive but benefits immensely from client-side acceleration via WebGPU, allowing it to run smoothly even on mid-range hardware [225]. Once the eyes are located, the next critical step is the precise localization of the pupil's center [222]. While early methods relied on simple heuristics like finding the darkest spot in the iris region, modern approaches leverage deep learning. Models like MediaPipe Iris or custom-built CNNs can accurately segment the pupil from the surrounding sclera, even under challenging lighting conditions or when partially obscured by eyelids [220][222]. The output of this stage is a pixel coordinate representing the center of the pupil within the captured image of the eye.

However, knowing the pupil's position relative to the eyeball is insufficient for determining where a person is looking on a screen. Head movements introduce significant noise into the calculation. A person can turn their head while keeping their gaze fixed straight ahead, or they can move their eyes to look sideways while keeping their head still. To disambiguate these cases, the system must simultaneously estimate the three-dimensional orientation of the head relative to the camera, known as head pose estimation [38] [560]. Using the 3D coordinates of the facial landmarks detected by MediaPipe, the system can calculate the Euler angles for pitch (up-down tilt), yaw (left-right rotation), and roll (tilting of the ear towards the shoulder) [564]. This head pose information is used to normalize the raw eye orientation data, correcting for the effect of head movement and isolating the true gaze direction of the eyeball itself. This fusion of head pose and eye orientation data is a common technique in appearance-based gaze estimation systems [329][401]. The final piece of the puzzle is calibration, a crucial procedure required to map the normalized eye and head orientation data to a specific point-of-gaze (PoG) on the computer screen [2] [6]. Without calibration, the system knows the direction of the gaze vector but not where it intersects the screen [560]. The most common calibration method involves presenting the user with a series of targets at known locations on the screen and recording the corresponding eye/head data while they fixate

on each target [198480]. The system then uses this paired data to train a regression model (e.g., weighted least squares) that learns the mapping from sensor inputs to screen coordinates [328333]. To reduce friction, research into calibration-free methods is ongoing, exploring techniques that use appearance-based features or analyze natural fixation patterns to establish a baseline mapping automatically [133306553]. For a high-stakes application like an interview, however, a brief, guided calibration step is essential for accuracy. During this phase, the system also records baseline metrics such as resting pupil size and skin tone under the ambient room lighting, which are vital for the subsequent PLR analysis [402].

The following table details the technical specifications and considerations for implementing the gaze tracking component.

Stage	Description	Key Technologies / Algorithms	Output
Face & Eye Detection	Detects the face and isolates the eye regions using facial landmarks.	MediaPipe Face Mesh [4], OpenCV, Haar Cascades [444]	468 facial landmarks; Cropped eye images
Pupil Center Localization	Determines the precise center of the pupil within the eye image.	Deep Learning (MediaPipe Iris [222]), Thresholding [222]	Pixel coordinates of the pupil center
Head Pose Estimation	Calculates the 3D orientation (pitch, yaw, roll) of the user's head.	Perspective-n-Point (PnP) algorithm using facial landmarks [190]	Euler angles (Pitch, Yaw, Roll)
Calibration	Maps sensor data (pupil, head pose) to screen coordinates (Point-of-Gaze).	Regression models (Weighted Least Squares [328]) on data from user-target interactions [6]	Calibration matrix/ model
Real-Time Inference	Continuously processes video frames to generate a stream of gaze data.	Custom CNNs, WebGPU-accelerated TensorFlow.js [91 225]	Time-series of gaze vectors, fixations, blinks

Once calibrated, the system enters its real-time operational mode. On each new frame from the webcam, the pipeline executes sequentially: face and eye detection, pupil localization, and head pose estimation. The resulting data is fed into the calibration model to compute the current point-of-gaze. This entire process must complete within a few milliseconds to maintain a smooth experience and meet the <200ms latency requirement . The final output is a rich, time-stamped data stream containing not only the gaze vector but also ancillary information like blink status and head pose, which is invaluable for context-aware analysis and anti-spoofing [275]. This continuous stream of behavioral data forms the first half of the dual-modal input, ready to be fused with the physiological data from the pupillary light reflex analysis module.

Implementation of Pupillary Light Reflex Analysis

The second mandatory modality, pupillary light reflex (PLR) analysis, provides the physiological layer of the dual-system, measuring involuntary, subconscious responses to visual stimuli [268]. Unlike gaze tracking, which measures voluntary behavior, PLR is governed by the autonomic nervous system and is extremely difficult to consciously control or fake [30 516]. This makes it an exceptionally powerful indicator of genuine engagement with the screen content. The implementation of PLR analysis requires a tightly coupled system of stimulus presentation and high-frequency response measurement, both executed within the client-side browser environment. The goal is to programmatically induce the reflex and then capture the subtle changes in pupil diameter with sufficient precision to extract meaningful diagnostic parameters [21 95]. This component directly addresses the challenge of verifying that a candidate's eyes are genuinely interacting with the digital display, forming a crucial check against sophisticated spoofing attacks.

The technical workflow for PLR analysis begins with the controlled presentation of a visual stimulus designed to elicit the reflex [21]. The PLR is triggered by changes in light intensity, so the system must dynamically alter the brightness of the screen [22]. This can be accomplished in several ways. A simple yet effective method is to rapidly flash the entire screen between a bright white state and a dark state, creating a clear, repeatable stimulus cycle . More advanced implementations might involve changing the color or intensity of specific UI elements, such as buttons or question fields, to create localized stimuli . The timing and properties of these stimuli (e.g., duration, intensity) must be logged with high precision, often synchronized with the system clock, as PLR parameters like latency are measured relative to the exact moment of stimulus onset [245520]. The accuracy of this stimulus presentation is paramount, as errors can invalidate the subsequent analysis. The system should aim for stimulus changes that are rapid enough to reliably trigger the reflex but not so fast as to cause visual discomfort or flicker fusion. Standardized guidelines for stimulus calibration, similar to those used in ophthalmic instruments (e.g., ISCEV guidelines), can inform the design of these visual cues [534540].

Concurrent with the stimulus presentation, the system must measure the pupil's response with high temporal resolution. This is achieved using the same webcam and computer vision pipeline as the gaze tracker. After the eye region is detected and the pupil localized, the system tracks the radius of the pupil over time [86]. Libraries like `jeelizPupillometry` demonstrate the feasibility of running such real-time pupillometry in the browser, leveraging the GPU for efficient computation [86]. To capture the dynamics of the PLR, which includes a relatively fast constriction phase and a slower

recovery phase, a high sampling rate is necessary [170][513]. The raw time-series data of pupil diameter will be noisy, containing artifacts from head movement, blinking, and minor shifts in lighting. Therefore, a preprocessing and signal processing stage is essential. This involves filtering the signal to remove noise and then applying algorithms to identify key features of the PLR waveform [521]. The primary parameters extracted from this analysis include:

- **Latency:** The time delay between the onset of the light stimulus and the start of pupil constriction. This is a fundamental characteristic of the reflex pathway and is typically around 200-300 milliseconds in healthy individuals [207][209][317].
- **Constriction Velocity:** The maximum speed at which the pupil diameter decreases during the initial response.
- **Amplitude:** The maximum change in pupil diameter from its baseline size to its fully constricted state [552].
- **Recovery Time:** The time it takes for the pupil to return to its baseline size after the light stimulus is removed [518].

These parameters form a quantitative profile of the subject's pupillary response. Deviations from expected values—for instance, an abnormally long latency or reduced constriction velocity—can indicate neurological issues or, in the context of proctoring, an unnatural state where the pupil is not reacting as it should [208][519]. The ability to measure these parameters remotely using a standard webcam has been demonstrated in several studies, validating the core premise of this component [96] [430]. The integration of PLR analysis with the gaze tracker creates a powerful synergy. While the gaze tracker confirms *where* the candidate is looking, the PLR analysis confirms that their eyes are physiologically *reacting* to what they see on the screen.

The following table summarizes the technical aspects of the PLR analysis implementation.

Aspect	Description	Implementation Details
Stimulus Control	Programmatically alters screen brightness/ intensity to elicit the PLR.	JavaScript-controlled canvas/screen flashes; Localized UI element changes .
Response Measurement	Tracks pupil diameter over time using the webcam.	Real-time pupillometry library (e.g., jeelizPupillometry [86]); High-frequency sampling (>60 Hz) [170].
Signal Processing	Filters raw data and extracts key PLR parameters.	Noise filtering; Algorithmic identification of waveform peaks/ troughs [521].
Key Parameters	Quantitative metrics derived from the PLR waveform.	Latency, Constriction Velocity, Amplitude, Recovery Time [317][518] [552].
Data Synchronization	Aligns PLR measurements with stimulus onset timestamps.	High-precision system clock synchronization; Timestampped log entries [245].

By combining precise stimulus control with high-fidelity response measurement, the PLR analysis module adds a critical layer of physiological verification to the proctoring system. It moves beyond observing surface-level behavior (gaze) to probing the underlying biological response, making it significantly more difficult for a candidate to convincingly simulate being engaged with the interview. The output of this module—a time-series of pupil diameter measurements correlated with screen events—is the second critical input stream for the data fusion engine, which will use it to validate or refute the signals coming from the gaze tracker.

Integrated Fusion Logic for Anomaly Detection and Liveness Verification

The true power and mandatory nature of the dual-modal system are realized not in the individual capabilities of gaze tracking and pupillary light reflex (PLR) analysis, but in their intelligent fusion. This integrated logic forms the analytical core of the system, correlating behavioral and physiological signals to detect anomalies indicative of cheating or spoofing. The primary enforcement mechanism is a correlation-based anomaly detector that looks for statistically improbable combinations of data. The most salient threat scenario—looking away from the screen to consult unauthorized materials—creates a distinct signature that the fused system is uniquely equipped to identify: a prolonged off-screen gaze accompanied by an absent pupillary light reflex to sustained screen brightness [9] [586]. By requiring both modalities to be active and continuously cross-validating their outputs, the system creates a defense-in-depth architecture that is far more robust than either modality could be alone. This fusion logic is what elevates the system from a collection of sensors to a coherent, intelligent proctor.

The foundation of this fusion logic is the establishment of a personalized baseline for each candidate during the initial setup and calibration phase . This baseline captures the individual's unique physiological and behavioral characteristics under normal, honest conditions. The system records several key metrics: the candidate's average resting pupil diameter under the ambient room lighting, their typical PLR latency and constriction velocity when presented with a standardized light stimulus, and their general gaze patterns when focusing on neutral content like the interviewer's video feed [206] [402]. These baselines are not static; they are established for the specific environmental conditions of the interview, particularly the lighting, which is a critical variable for both pupil size and

PLR measurement [5][19][5][31]. With this baseline established, the system can begin real-time analysis, flagging any significant deviation from the norm as a potential risk event.

During the interview, the fusion engine continuously monitors the two data streams. The primary pattern it hunts for is the "off-screen gaze + absent PLR" signature. The gaze tracker reports the direction of the candidate's gaze. If this vector indicates the candidate is looking off-screen (e.g., to the top-right corner, a common location for hidden notes) for a duration exceeding a set threshold (e.g., 3 seconds), a low-level alert is generated [9] . However, this by itself is not conclusive evidence of cheating, as a candidate might legitimately look away to think. Here is where the PLR module provides the decisive confirmation. The PLR analyzer is constantly monitoring the pupil diameter and the screen's luminance. If the candidate's gaze is off-screen, but the screen's background remains brightly lit, the system expects to see a PLR—the pupil should constrict in response to the light. If the system observes that the pupil diameter remains large and shows no signs of constriction, it constitutes strong corroborating evidence. The absence of a physiological response when a response is expected is a powerful indicator of an unnatural state, suggesting the candidate is not actually looking at the screen [26][82][69]. The simultaneous occurrence of these two events—an off-screen gaze from the behavioral modality and an absent PLR from the physiological modality—triggers a high-confidence violation. The system can then escalate this event to a recruiter for review or, according to predefined policies, take automated actions like pausing the interview or terminating the session .

Beyond detecting cheating, this dual-modality system is exceptionally effective for liveness detection, a critical component of anti-spoofing [14][2]. Spoofing attacks, such as presenting a photograph, a recorded video, or a high-quality 3D mask of another person, rely on static or pre-recorded representations. These representations inherently lack the dynamic biological properties of a living person. The PLR analysis provides a direct countermeasure. A static photo or a simple video replay will not exhibit a pupillary light reflex; the pupil in the image will remain fixed regardless of changes in screen brightness [27][2]. Furthermore, active liveness checks can be integrated into the workflow. The system can randomly prompt the candidate to follow a moving dot on the screen with their eyes . A successful response requires both valid gaze tracking (the eyes must move to follow the dot) and a corresponding PLR (the pupil should react to the light changes inherent in the animation). A 3D mask might replicate a static gaze but may struggle to perfectly mimic the subtle micro-movements (saccades, blinks) and the full dynamic range of a real PLR, especially under inconsistent lighting [26][63][48]. The combination of passive, continuous monitoring (correlating gaze with PLR) and periodic active challenges (requiring the candidate to perform specific eye movements) creates multiple layers of defense against a

wide spectrum of spoofing attacks, from basic photo replays to more advanced deepfakes [272][432]. Compliance with international standards like ISO/IEC 30107-3, which defines methodologies for testing Presentation Attack Detection (PAD), should be a key design goal to ensure the system's anti-spoofing capabilities are robust and verifiable [176][177][283]. The fusion of gaze and PLR, therefore, serves a dual purpose: it detects intent to cheat and it verifies the physical presence of the authentic candidate.

The following table outlines the logic for the integrated anomaly detection system.

Anomalous Pattern	Behavioral Modality (Gaze)	Physiological Modality (PLR)	System Interpretation & Action
Potential Cheating	Gaze vector indicates off-screen look for >3s.	Pupil remains dilated despite bright screen background.	**High Confidence Violation.** Candidate is likely viewing unauthorized aids. Trigger high-priority flag. [9] [586]
Possible Distraction	Gaze vector indicates off-screen look for <3s.	PLR is present but may be weaker or delayed.	**Medium Priority Flag.** Candidate may be thinking or glancing away. Log for reviewer consideration.
Spoofing Attempt (Photo)	Gaze tracking is unstable or fails.	No PLR is detected when screen brightness changes.	**Critical Violation.** Evidence of a static image. Immediately terminate session. [272][346]
Spoofing Attempt (Video)	Gaze tracking may be inaccurate or laggy.	Absent or irregular PLR; lacks natural dynamics.	**Critical Violation.** Evidence of a video replay. Immediately terminate session. [272]
Spoofing Attempt (Mask)	Gaze may be present but lacks natural micro-movements.	PLR may be present but has abnormal latency/velocity.	**High Confidence Violation.** Evidence of a sophisticated mask. Trigger high-priority flag. [268] [269]
Normal Behavior	Gaze vector corresponds with on-screen activity.	PLR is present and correlates with screen brightness changes.	**No Action.** Baseline condition confirmed. Continue monitoring.

This sophisticated fusion logic, grounded in the principles of multimodal biometrics, is what makes the system truly mandatory and effective. It doesn't just collect data; it interprets it contextually, creating a holistic view of the candidate's state. By enforcing a correlation between what the candidate's eyes are doing and what their pupils are feeling, the system closes the loopholes that exist in single-modality systems, delivering a level of integrity assurance that is essential for high-stakes remote interviews.

System Architecture, Data Management, and Reporting

The successful implementation of a mandatory dual-modal proctoring system requires a robust, scalable, and secure system architecture that seamlessly integrates the client-side monitoring with a powerful backend for data aggregation, analysis, and reporting. The

proposed Hybrid Edge-Cloud model is central to this design, leveraging the strengths of both environments: the low-latency, real-time processing capabilities of the client's browser for inference, and the centralized, high-throughput processing power of the cloud for session management, telemetry ingestion, and long-term data storage . This division of labor ensures that the most time-sensitive tasks are handled locally to avoid streaming delays, while the heavy lifting of data analysis and reporting occurs in a controlled server-side environment. The architecture must support the continuous, high-frequency streaming of biometric data from thousands of concurrent users, demanding a carefully designed data pipeline and a polyglot persistence strategy to handle the diverse data types involved.

On the client side, the application is responsible for orchestrating the entire proctoring workflow. It initializes the camera feed using WebRTC for low-latency streaming . The core of the client-side application is the WebAssembly/WebGPU-powered inference engine, which runs the CV models for gaze and PLR analysis in real-time [59] [61]. This engine continuously processes video frames and generates a stream of telemetry data. To manage network bandwidth and latency, this data is serialized into compact JSON payloads and sent to the backend via a dedicated WebSocket endpoint, `/ws/v1/telemetry` . These payloads contain timestamped data points for gaze vectors, head pose, and pupil diameter, compressed using an efficient format like Protobuf or MessagePack to keep the data rate under 50 KB/s . The client application also manages the UI flow, guiding the user through the mandatory onboarding, calibration, and interview phases, and communicating system status and warnings back to the candidate .

The backend architecture is designed for scalability and reliability. At its heart is an API Gateway (using GraphQL for flexible data fetching and REST for webhooks) that routes incoming requests . Real-time telemetry from clients is ingested by a high-throughput messaging system like Apache Kafka, which decouples the data producers (clients) from the data consumers (processing services) and buffers the massive influx of data . This allows the system to handle traffic spikes gracefully. Processing Workers, built with Python and FastAPI, consume the telemetry streams from Kafka. These microservices are responsible for a variety of tasks: writing the raw time-series data to a specialized database, running more computationally intensive anomaly detection algorithms that may not be suitable for the client-side, and aggregating data to calculate the final Integrity Score . The Recruiter Dashboard is also served by the backend, which fetches aggregated session data and risk events to populate its timeline and scoring interface. For scenarios where client-side processing is too resource-intensive (e.g., older hardware), the architecture must have a fallback to cloud-based processing, although this introduces higher latency and should be avoided when possible . The entire backend is deployed on

a Kubernetes cluster with auto-scaling to ensure 99.99% uptime, meeting enterprise-grade availability expectations .

Effective data management is critical for both performance and regulatory compliance. Given the heterogeneous nature of the data—from high-frequency time-series telemetry to structured metadata and large video files—a polyglot persistence approach is necessary . The following table details the recommended database choices:

Data Type	Database Technology	Reasoning
Biometric Telemetry	TimescaleDB (PostgreSQL extension) or InfluxDB	Optimized for high-write throughput of time-series data; enables efficient queries over time ranges.
User & Session Metadata	PostgreSQL	ACID-compliant relational database for structured data like user profiles, session status, and permissions.
Facial Embeddings	Vector Database (e.g., Pinecone, Milvus)	Designed for efficient similarity search and storage of high-dimensional numerical vectors used for identity verification.
Interview Artifacts (Video/Audio)	AWS S3 (Object Storage)	Scalable, durable, and cost-effective storage for large binary files, with fine-grained access control.

This schema design supports the end-to-end application flow. The backend receives a continuous stream of biometric data, which is stored in the hypertable `biometric_telemetry`. When a risk event is detected—for example, an extended off-screen gaze—it is inserted into the `risk_events` table with a timestamp, event type, and confidence score, along with any relevant metadata . After the session ends, a backend job aggregates all the telemetry and risk events associated with a `session_id` to compute the final `integrity_score`, which is then saved in the `sessions` table . This structured data is the foundation for the recruiter's analytics dashboard.

The recruiter-facing interface, or "Command Center," is the culmination of this architectural effort. It provides a comprehensive, multi-layered view of the interview session. The main display is a video player showing the recorded interview . To the side, a "Ghost View" overlay can be toggled on, drawing the skeletal wireframe and gaze vectors over the candidate's head, giving the proctor a direct visualization of what the AI saw . Below the video, a multi-layered timeline displays different streams of telemetry: a gaze heatmap showing where the candidate looked, a graph of head pose angles, and a spectrogram of audio anomalies . Clicking a red "Risk Flag" marker on this timeline instantly jumps the video to that precise moment and pauses it, while a side panel displays the underlying telemetry data that triggered the flag, such as "Gaze vector detected looking at secondary screen for 4.2 seconds" . This rich, interactive reporting transforms raw data into actionable intelligence, empowering recruiters to make informed decisions about candidate integrity. The entire process, from real-time

monitoring on the client to post-session forensic analysis in the cloud, is supported by this robust and scalable architecture.

Security, Privacy, and Compliance Framework

Implementing a dual-modal biometric proctoring system necessitates a rigorous security and privacy framework, given the highly sensitive nature of the collected data. Biometric information, including facial geometry, eye movements, and pupillary responses, is considered Personally Identifiable Information (PII) and is protected under numerous global regulations such as the General Data Protection Regulation (GDPR) in Europe, the Biometric Information Privacy Act (BIPA) in Illinois, and the California Consumer Privacy Act (CCPA) [422]. The system's design must therefore embed privacy and security by default, adhering to a "Privacy-Protecting Techniques for Behavioral Biometric Data" survey [54]. The architecture must minimize data exposure, employ strong encryption, and provide transparency and control to the user, all while upholding the highest industry standards for authentication and anti-spoofing.

The core principle of data security is end-to-end protection. All biometric data must be encrypted both in transit and at rest . Communication between the client browser and the cloud backend must use TLS 1.3 to prevent eavesdropping . For data at rest, all databases and storage systems must be encrypted using strong algorithms like AES-256 . A particularly critical aspect is the handling of raw biometric templates, such as the candidate's facial embedding vector used for identity verification. Instead of storing these vectors in plaintext, the system should store only a secure cryptographic hash of the vector in the relational database [215]. This ensures that even if the database is compromised, the original biometric template cannot be easily reconstructed. Raw video recordings, which contain the most sensitive data, should ideally never leave the client's device unless a high-confidence risk event is detected. Even then, access to these recordings should be strictly controlled, granted only via short-lived, pre-signed URLs and audited meticulously . Furthermore, all Personally Identifiable Information (PII) within the application, such as candidate names and emails, should be masked or tokenized in analytics dashboards and logs to prevent accidental exposure [2] [21].

In addition to data protection, the system must comply with recognized industry standards for biometric authentication and anti-spoofing. For authentication assurance levels, the system's design should align with the NIST SP 800-63 Digital Identity Guidelines [490] [542]. Specifically, NIST SP 800-63B recommends the use of liveness

detection for any biometric authentication system intended for high-assurance identity proofing, a clear mandate for the PLR and active gaze-tracking components of this system [162588]. The system should be architected to support evolving NIST guidance, such as the emphasis on phishing-resistant MFA and continuous, in-session evaluation of user and device signals to catch fraud after login [434574575]. For anti-spoofing, compliance with the ISO/IEC 30107 series of standards is a key benchmark [176]. The standard ISO/IEC 30107-3 specifically establishes methodologies for assessing the performance of Presentation Attack Detection (PAD) mechanisms [177286]. Achieving a compliance level like iBeta Level 2 would demonstrate that the system's defenses against photo, video, and print attacks have been independently tested and validated against a rigorous suite of attack types [178283]. Adherence to these standards is not just a best practice; it is becoming a market expectation and a prerequisite for enterprise adoption [283].

Finally, the system must incorporate ethical safeguards and provide recourse for users. The TRD's mention of an "Accessibility and Usability" section highlights a commitment to responsible AI . The system should offer alternative flows for candidates with documented medical conditions that affect eye movement or pupil response, such as disabling strict gaze tracking upon request . Consent is a cornerstone of biometric privacy. Candidates must provide explicit, informed consent before any data is collected, with clear explanations of what is being monitored and why [127]. This consent mechanism should be granular and revocable. The system should also be designed to mitigate algorithmic bias. The training datasets for the CV models must be diverse and representative of different demographics, including various skin tones, genders, and ages, to ensure the system performs equitably across all user groups [130567]. Regular fairness audits should be conducted to identify and correct any disparities in false positive rates. By integrating these security, compliance, and ethical considerations directly into the system's architecture and development lifecycle, the AI Interview Room can build trust and provide a solution that is not only technologically advanced but also socially responsible and legally compliant.

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